

# Nusret Efe Ucer

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## EDUCATION

### THE UNIVERSITY OF CHICAGO

Chicago, IL

Bachelor of Science in Computer Science, Bachelor of Arts in Economics

Expected: June 2028

**GPA:** 3.82/ 4.00

**Awards:** UChicago 2025 Summer Tech Showcase Runner-up; UChicago 2025 Winter Tech Showcase; UChicago Grant for Hackathon; Jeff Metcalf Award Recipient x2; IMC Prosperity 4 Finalist

**Relevant Coursework:** Multiscale Modeling of Biological Systems I & II, Honors Discrete Mathematics, TTIC31150 Graduate Abstract Linear Algebra and Probability, Introduction to Database Systems, Theory of Algorithms, Machine Learning, Statistical Methods and Applications, Investments, Elements of Economic Analysis I, Introduction to Computer Security

## TECHNICAL SKILLS

**Languages:** Python, C, C++, Rust, Java, R, JavaScript, HTML/CSS, Tcl/Tk

**Frameworks/Tools:** Flask, Node.js, Nest.js, Streamlit, Git, Azure, GCP, Firebase

**Databases/Platforms:** Azure SQL & Databricks, MongoDB, Elasticsearch, OpenAI APIs, Vertex AI

## EXPERIENCE

### Velocity Catalyst Fund

Chicago, IL

*Incoming Summer 2026 Pipeline Tool Management Intern*

June 2026 – August 2026

### G LNK YC W25

Chicago, IL

*ML Engineer (Part-Time) / ML Engineering Intern*

June 2025 – Present

- **Built the company's database from zero**, scraping and unifying 300+ government healthcare datasets across CMS, NHS, and the French Ministry of Health; transformed  $10^9+$  Azure SQL records into  $10^7$ -scale Elasticsearch, Databricks, and MongoDB systems with custom schemas, indexing, ownership hierarchies, and cross-dataset links using OpenAI and Vertex AI.
- **Engineered the client-facing ETL pipeline** for non-healthcare expansion and data enrichment; optimized credit usage and parallelized execution across Azure VMs, multithreading, and asynchronous workflows, enabling same-day integrations and scaling coverage to approximately 92% of the global market across 170+ countries.
- **Built a full-stack search and analytics system** with Elasticsearch- and MongoDB-backed search, multi-criteria and regex filtering, a DataGrid frontend, and analysis pages with computed metrics and cost breakdowns.

### UChicago Polarization-modulated Laser Satellite Experiment (PULSE-A)

Chicago, IL

*Ground Station Software Engineer*

October 2024 – Present

- **Built the PyQt Ground Station control system from scratch**, integrating GUI, backend logic, multithreaded and asynchronous process management, and real-time command and data handling for the Optical Ground Station telescope and laser-communication assembly.
- **Developed communication workflows** for a fully student-led satellite launch, designing latency-aware protocols for orbital tracking and reliable low-latency inter-city communication between Optical and RF Ground Stations.

## TECHNICAL PROJECTS

### Maroon Exchange

June 2025 – August 2025

- **Co-built a peer-to-peer borrowing and lending platform** for University of Chicago students using React, Firebase authentication, and Azure SQL, with notification-driven matching, lender portfolios, and post-transaction ratings; runner-up at the UChicago Summer Tech Showcase, awarded \$5K.

### “HandsIn” — HackTech 2025

April 2025

- **Built a sign-language conversational system** at Caltech HackTech after winning school grant funding; created a custom ASL dataset, trained a neural network from scratch, connected signing input to Gemini, and reconstructed responses in ASL through DeepMotion for a real-time signing avatar.

### R19Q Mutation Analysis

March 2025 – May 2025

- **Analyzed the R19Q mutation in the SBDS protein** using a 20 ns molecular dynamics simulation on Midway3; performed RMSD/RMSF, Kabsch alignment, SASA, radius-of-gyration, hydrogen-bond, and time-series analyses to demonstrate mutation-driven domain dissociation associated with Shwachman–Diamond syndrome.

## LEADERSHIP AND ACTIVITIES

### UChicago Association for Computer Machinery

Chicago, IL

*President*

October 2024 – Present

- **Led the AI/ML committee for ACM**, organizing events, workshops, and journal discussions on AI/ML literature, LLM fundamentals, model-building, prompt engineering, jailbreaking, and hyperparameter tuning.
- **Served as a core ACM board member**, overseeing event organization, interviewing new members, and participating in board meetings supporting chapter operations.